

Di Wu

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EDUCATION

• University of Science and Technology of China [🌐]

B.S. in Applied Physics

GPA: 3.91/4.30

Hefei, China

Sep. 2021 - July 2025

◦ Relevant Coursework :

Computer Architecture H	93/100	Computer Organization	96/100
Computer Programming A	95/100	Computational Method	95/100
Computer Network	87/100	Operation System	85/100
Linear Algebra	100/100	Artificial Intelligence Foundation	93/100
Probability Theory & Math. Statistics	96/100	Mathematical Analysis	93/100

• University of Toronto[🌐]

Ph.D. in Computer Science, supervised by Prof. [Qizhen Zhang](#)

Toronto, Canada

Sep. 2025 - Dec. 2030 (expected)

RESEARCH EXPERIENCE

• Chair, the System and Networking Group, UofT

Apr. 2025 - Present

Supervisor: Prof. [Qizhen Zhang](#)

Project I: *Massive Communication Framework in GPU for Database*

- Conducted analysis of current GPU communication frameworks, and explored advanced technique.
- Designed the new GPU communication framework for DB with high-performance and optimization.
- Accepted by VLDB'26.

Project II: *GPU Database Benchmark*

- Conducted detailed analysis of famous public GPU databases, and provided insights of the bottleneck.

Project III: *Database Agent System*

• Chair, the System and Networking Group, NUS

July 2024 - Jan. 2025

Supervisor: Prof. [Bingsheng He](#) and Prof. [Yao Chen](#)

Project I: *Content Addressable Memory on FPGA*

- Conducted analysis of FPGA-based CAM, enhancing the theoretical foundation for this DSA design.
- Proposed a configurable DSP-based CAM architecture with high-performance and multi-query support.
- Implemented an extensible CAM kernel with a hybrid HLS and RTL architecture on FPGA.

• Research Assistant, Institute of ASIC, USTC

June 2023 - June 2024

Supervisor: Prof. [Lei Zhao](#)

Project I: *Test and analysis of a SerDes prototype chip based on 180 nm CMOS process*

- Investigated chip configurations and built test code using Verilog HDL.
- Utilized FPGA to test the relevant index shortness, providing data to enhance the feed-through design.

Project II: *Simulation of the chip wiring S parameters*

- Built the simulation models for chip wiring based on the encapsulated drawings.
- Determined the S parameters providing important information for the post-simulation circuit analysis.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, O=ORAL PRESENTATION, S=IN SUBMISSION, T=THESIS

- [C.1] Di Wu, Hongshi Tan, Hanzhang Yang, Bingsheng He, Qizhen Zhang. "MGI: A Communication Framework for Data Processing in Massive GPU Infrastructures," in 2026 VLDB. [🌐]
- [C.1] Yao Chen, Feng Yu, Di Wu, Weng-Fai Wong and Bingsheng He. "Configurable DSP-Based CAM Architecture for Data-Intensive Applications on FPGAs," in 2025 62th Design Automation Conference (DAC). [🌐]
- [O.1] Jiacheng Guo, Jiajun Qin, Lei Zhao, Wu Di, Xinyu Bin, "A 4×6.25-Gbs Serial Link Transmitter Core in 0.18-μm CMOS for high-speed front-end ASICs," in 2024 24th IEEE Real Time Conference.
- [J.1] Yuheng Wu, Xiangdong Kong, Yechao Su, Jiankang Zhao, Yiling Ma, Tongzheng Ji, Di Wu, Junyang Meng, Yan Liu, Zhigang Geng, Jie Zeng, "Thiol Ligand-Modified Au for Highly Efficient Electroreduction of Nitrate to Ammonia," in *Precis. Chem.* 2024, 2, 3, 112–119.

TA EXPERIENCE

- **University of Toronto**

CSC258H1F Computer Organization

CSC367H1S Parallel Programming

Sep. 2025 – Dec. 2025

Jan. 2026 – Apr. 2026

SKILLS & INTERESTS

- **Programming Languages:** C/C++, Verilog/System Verilog, Python, L^AT_EX
- **Developer Tools:** Vivado, Vitis, Nsight
- **Technologies/Frameworks:** Linux, Git, HLS, CUDA, Pytorch
- **Interests:** Students Chorus Member, Piano